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Education

Babu Banarasi Das University

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Lucknow, UP

Expected May 2027

- **Relevant Coursework:** Machine Learning, Data Science, Artificial Neural Networks, Computer Networking, Database Management Systems, Design and Analysis of Algorithms.

IIT Mandi

Minor in Artificial Intelligence

Mandi, HP

March 2025 – Present

- **Relevant Coursework:** Programming in Python, Linear Algebra, Calculus, Statistics, Probability, Traditional AI and Machine Learning, Deep Learning for Computer Vision

Projects

Cognova: Amazon ML Challenge '25 - Multimodal Price Prediction | *LightGBM, XGBoost, Optuna*

- Achieved a competitive 58.02 SMAPE score in multimodal price prediction by architecting a robust dual-pipeline LightGBM and XGBoost ensemble.
- Improved prediction stability across a 5-fold out-of-fold (OOF) generation by extracting text via Sentence-BERT and processing images via ResNet-50, caching 2048-dimensional features as FP16 memmaps.
- Resolved OOF log-scale mismatches and maximized ensemble accuracy by automating hyperparameter tuning with Optuna and tracking artifacts meticulously via MLflow.

JAX Deep Dive: GPU Fraud Detection Network | *JAX, Python, CUDA, Functional ML*

- Engineered a GPU-accelerated fraud detection system handling a 577:1 class imbalance across 284,807 European credit card transactions by utilizing JAX functional programming paradigms.
- Achieved a 3.99x training speedup (2.30s vs 9.17s) over PyTorch while maintaining a 0.9536 ROC-AUC and 0.6471 PR-AUC by leveraging `@jax.jit` compilation and `vmap` batch transformations.
- Democratized advanced functional ML concepts by authoring a comprehensive, from-scratch JAX framework tutorial documenting automatic differentiation and GPU scaling.

nanoDist: Distributed Engine (In Development) | *NumPy, HPC, Distributed Systems*

- Maximizing memory efficiency across simulated distributed worker nodes by architecting a custom machine learning execution stack entirely in pure NumPy.
- Scaling foundational model training without external frameworks by engineering manual automatic differentiation and ring all-reduce communication primitives.
- Minimizing data flow and parameter memory footprints by implementing ZeRO-1 optimizer state partitioning and mixed-precision operations from scratch.

HiveTorch: Federated Core (In Development) | *PyTorch, Federated Learning, Statistics*

- Enabling privacy-preserving collaborative learning across edge devices by building an end-to-end decentralized Federated Averaging (FedAvg) pipeline in PyTorch.
- Evaluating statistical data heterogeneity and model convergence by implementing Dirichlet distribution sharding to simulate complex non-IID data topologies.
- Aggregating decentralized knowledge securely by orchestrating custom client-side SGD loops and centralized server-side state-dict aggregations.

LumaSort-Engine | *C++20, OpenGL Compute Shaders, OpenCV*

- Achieved real-time rendering speeds of 60+ FPS for complex visual transformations by developing a custom C++20 engine utilizing OpenGL Compute Shaders.
- Morphed live multimodal inputs (webcams, drawings, static images) into target shapes by implementing global luminance-based 1D optimal transport algorithms.
- Maximized particle physics performance and responsiveness by deploying massively parallel pixel processing and dynamic fluid dynamics with real-time tuning capabilities.

Technical Skills

Languages: Python, C, C++ (C++17/20), Go, SQL

Technologies:

– **Frameworks & ML:** PyTorch, JAX, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV

– **MLOps & Tooling:** Weights & Biases (wandb), MLflow, Optuna, Docker, Git, Linux

– **Backend & Systems:** gRPC, Redis, AVX2 SIMD Intrinsics, Memory Management, Distributed Data Parallel

Concepts: Artificial Intelligence, High-Performance Computing (HPC), Linear Algebra, System Design

Achievements

- Secured **2nd Place at HackArena 2025** for **CivicAgent**. Engineered the core AI routing and deterministic backend architecture to autonomously triage civic issues without manual intervention.

Social Engagements & Leadership

Campus Partner: Perplexity AI (Fall '25 Cohort, Aug 2025 – Jan 2026)

Member: Google Developer Groups (GDG) Lucknow (Aug 2025 – Present)